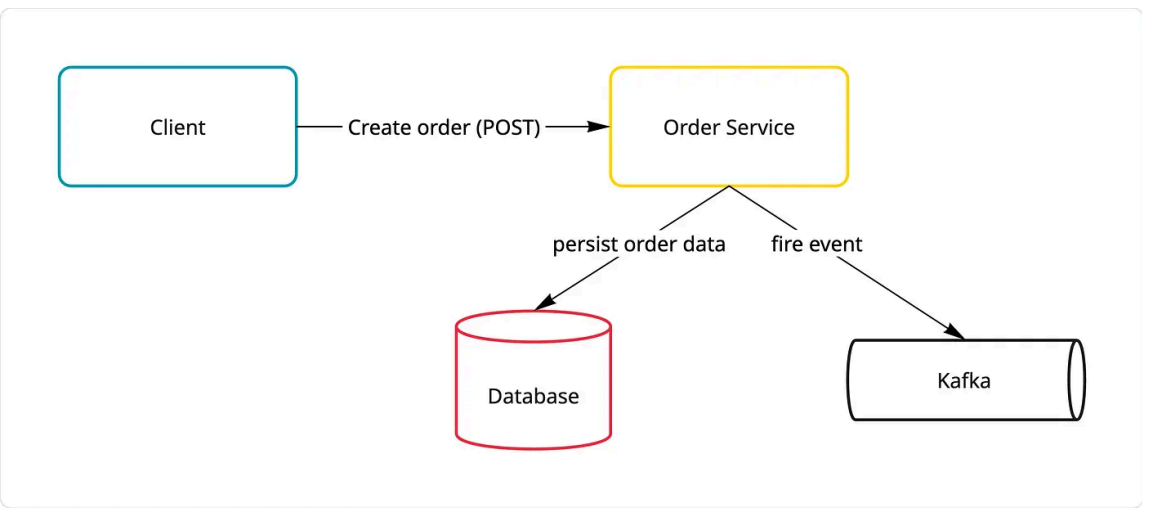
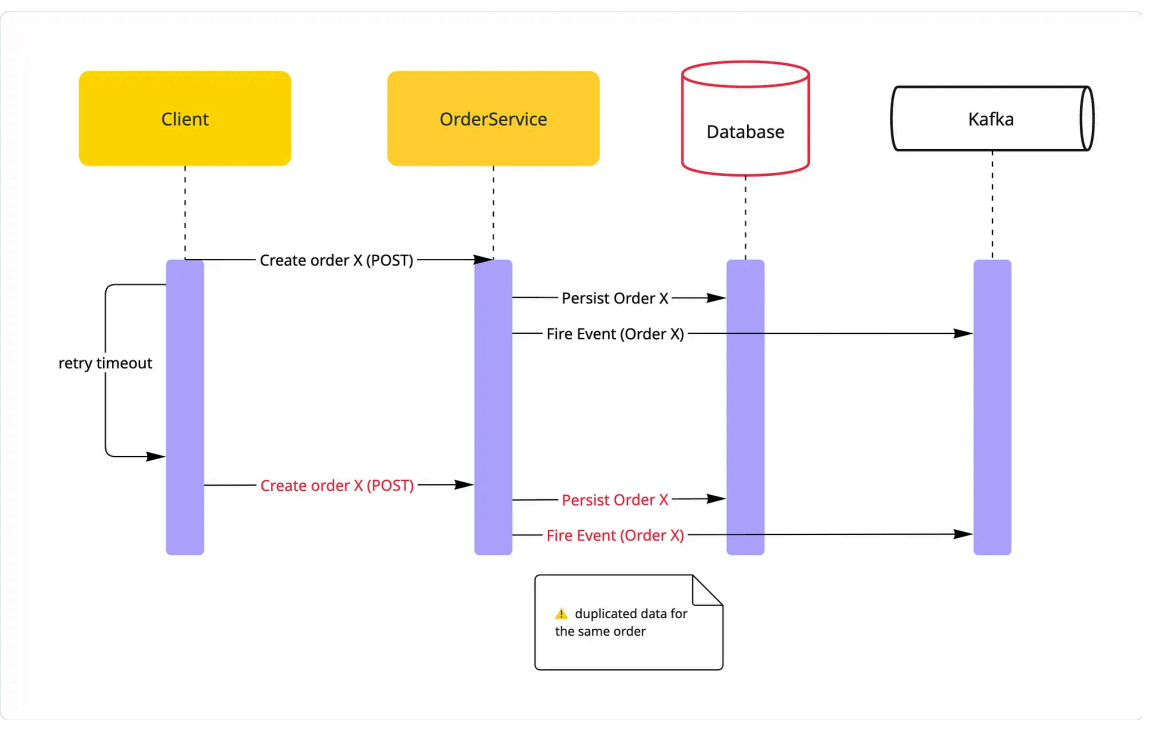
Implementing Idempotent POST Requests

- Idempotence = operation that can be repeated with the same effect
- GET/PUT/DELETE → idempotent
- POST → *not* idempotent
- Duplicate POSTs → duplicate orders / events


Client
Create order (POST)
Order Service
persist order data
fire event
Database
Kafka

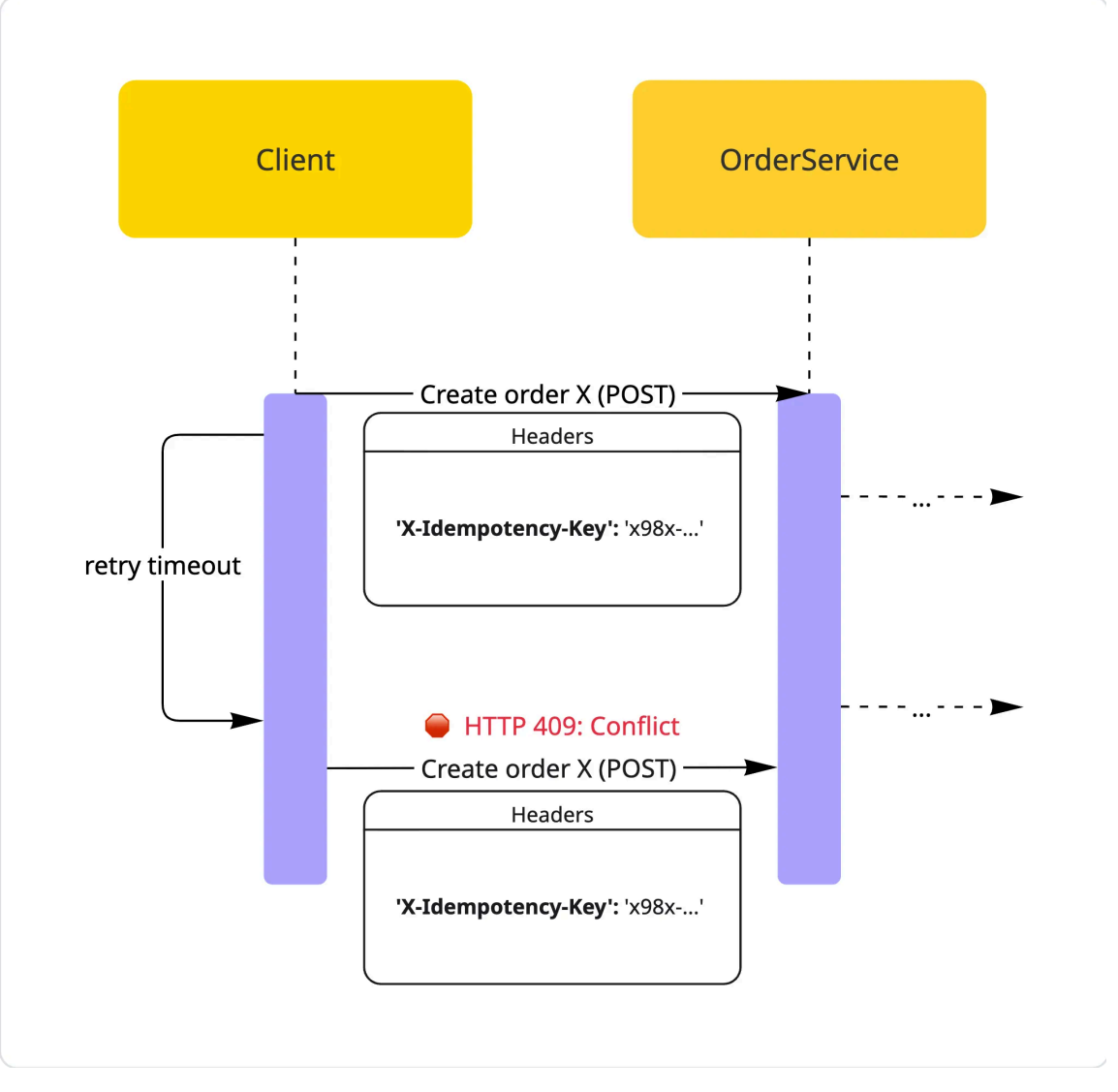
The Problem

- Retries happen because of:
  - client timeouts
  - network failures
  - server crashes
- POST /orders sent twice may create two orders
- DB constraints don't help when the same order is valid twice
- Downstream (Kafka, billing, logistics) → becomes a mess


Client
OrderService
Database
Kafka
Create order X (POST)
Persist Order X
Fire Event (Order X)
retry timeout
Create order X (POST)
Persist Order X
Fire Event (Order X)
⚠ duplicated data for the same order

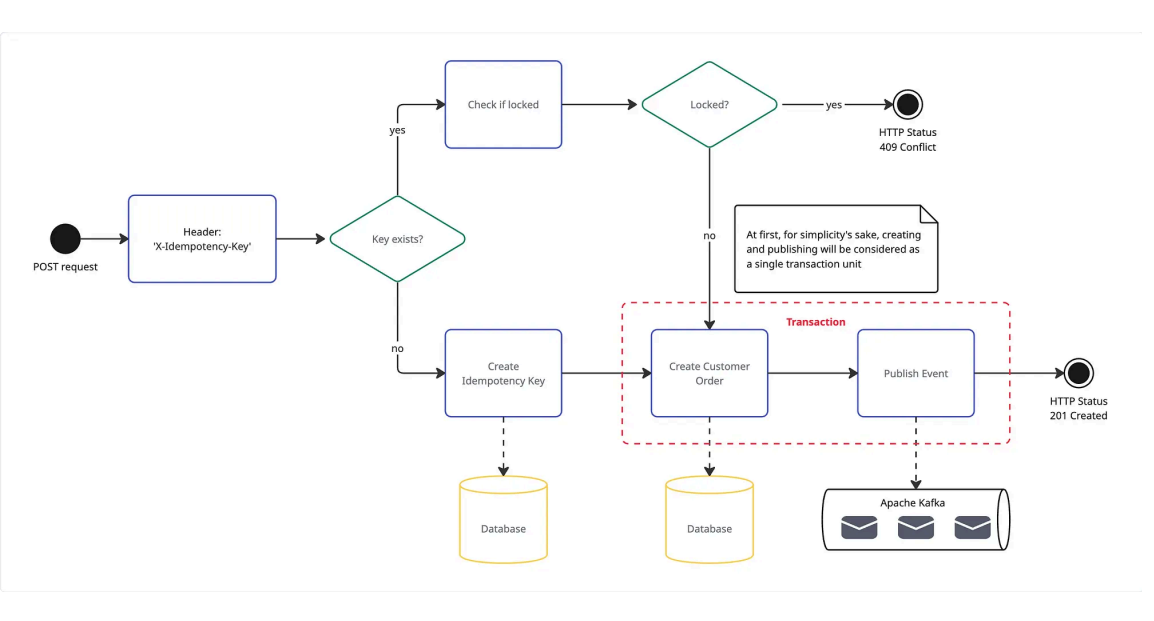
Solution Overview

- Client sends a **unique idempotency key**
  - Header: `X-Idempotency-Key: <uuid>`
- Server stores the key
- If key already present → immediately return **409 Conflict**
- Guarantee: each *intent* is processed once


Client
OrderService
Create order X (POST)
Headers
'X-Idempotency-Key': 'x98x-...'
retry timeout
HTTP 409: Conflict
Create order X (POST)
Headers
'X-Idempotency-Key': 'x98x-...'
...
...

Architecture

- Client → sends key + request body
- Server:
  1. try to create key record (transaction)
  2. if locked → 409
  3. else create order + publish event
  4. return 201


POST request
Header: 'X-Idempotency-Key'
Key exists?
yes
Check if locked
Locked?
yes
HTTP Status 409 Conflict
no
no
Create Idempotency Key
Database
At first, for simplicity's sake, creating and publishing will be considered as a single transaction unit
Transaction
Create Customer Order
Database
Publish Event
Apache Kafka
HTTP Status 201 Created

Controller Example

```
@PostMapping
public ResponseEntity<?> createOrder(
    @RequestHeader("X-Idempotency-Key") String key,
    @RequestBody CreateCustomerOrderDto order) {

    IdempotencyKey entity;
    try {
        entity = idempotencyKeyService
            .getOrCreateIdempotencyKeyByKeyAndUserId(key, order.getUserId());
    } catch (IllegalStateException e) {
        return ResponseEntity.status(HttpStatus.CONFLICT)
            .body(Map.of("status","duplicate"));
    }

    Long orderId = ordersService.createOrder(order);
    kafkaProducerService.sendMessage("orders-topic", orderId);

    return ResponseEntity.status(HttpStatus.CREATED)
        .body(Map.of("orderId", orderId));
}
```

Idempotency Service

```
@Transactional
public IdempotencyKey getOrCreateIdempotencyKeyByKeyAndUserId(
       String key, String userId) {

    Optional<IdempotencyKey> existing =
        repo.findByIdempotencyKeyAndUserId(key, userId);

    IdempotencyKey idKey;

    if (existing.isPresent()) {
        idKey = existing.get();
        if (checkIfIsLocked(idKey)) {
            throw new IllegalStateException("Key is locked");
        }
    } else {
        idKey = new IdempotencyKey(key, userId);
    }

    idKey.setLockedAt(LocalDateTime.now());
    return repo.save(idKey);
}
```

Resources